Densest Community Search

Query-anchored dense subgraphs in directed citation networks

The-Anh Vu-Le

University of Illinois Urbana-Champaign

vltanh@illinois.edu

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Motivation and problem statements

Motivation

- Citation networks: nodes are papers, directed edges are citations
- Query: a paper q
- Goal: retrieve a "good" citation community around q

Question. Can we search for a neighborhood that is dense, query-centered, and structurally constrained?

Oracle access

Edge convention: $u \to v \iff u$ cites v.

The graph is available only through 1-hop neighborhood queries. Each query for a paper v returns its outgoing references and its incoming citations.

Incoming-citation cap

- incoming-edge queries can be more expensive than outgoing:
  - reference list is usually small, cited-by list is unlimited
  - references read off the paper itself; cited-by needs a maintained database
- the cap limits how many incoming edges are fetched per queried node

Combinatorial problem

Given

- directed graph $G = (V, E)$
- query node $q \in V$
- minimum size $k \geq 3$

Solve

$$\begin{aligned} \max_{S \subseteq V} \quad & \frac{|E(S)|}{|S|(|S|-1)} \\ \text{s.t.} \quad & q \in S \\ & |S| \geq k \end{aligned}$$

where $E(S) = \{(u, v) \in E \mid u, v \in S\}$.

Optional connectivity

Rooted κ-edge-connectivity

For every selected node $v \in S \setminus \{q\}$:

there are at least κ edge-disjoint paths from q to v

in the underlying undirected graph.

Note

- $\kappa = 0$ disables the constraint.
- Enforced lazily by min-cut cuts inside the solver.

Approaches

Dinkelbach transform

For a density estimate λ, define

$$Z(\lambda) = \max_S \{|E(S)| - \lambda|S|(|S|-1)\}.$$

Property

$$\lambda^\star \text{ is optimal} \quad \Longleftrightarrow \quad Z(\lambda^\star) = 0.$$

- If $Z(\lambda) > 0$, an improving denser set exists
- Update $\lambda \leftarrow |E(S)|/[|S|(|S|-1)]$
- Stop when the parametric subproblem cannot improve density

Dinkelbach loop

1. **Initialize:** build a feasible set by BFS from q, prune to improve density, set λ_1 to its density, $t = 1$
2. **Optimize:** solve the parametric integer subproblem at λ_t
3. **Convergence check:** if $Z(\lambda_t) \leq 0$, terminate and return λ_t
4. **Update:** otherwise set λ_{t+1} to the density of the new subgraph, $t \leftarrow t + 1$, return to step 2

Outer loop handles the fractional objective.

Inner loop solves the parametric problem by branch-and-price.

See appendix A5 for the LP formulation and linearisation.

Active set and frontier

Maintained sets

- A: active nodes in the restricted master problem
- F: frontier nodes discovered but not active
- subgraph induced by $A \cup F$ is kept up to date via oracle queries, so no internal edge is missed

Initialization

- BFS rooted at q: each dequeued node is oracle-queried, joins A, and its neighbors enter F
- loop terminates at $|A| = k$

Column generation

Reduced cost of a frontier node $f \in F$ at RMP optimum $\bar{x}$ with size-constraint dual π:

$$\bar{c}_f = \underbrace{d_{\mathrm{frac}}(f)}_{\text{edge gain}} - \underbrace{2\lambda \sum_{v \in A} \bar{x}_v}_{\text{pair penalty}} - \underbrace{\pi}_{\text{size dual}}, \qquad d_{\mathrm{frac}}(f) = \sum_{v \in N(f) \cap A} \bar{x}_v.$$

Pick $f^\star = \operatorname*{argmax}_{f \in F} \bar{c}_f$. If $\bar{c}_{f^\star} \leq 0$, RMP is a valid upper bound and CG terminates. Otherwise promote the top-b frontier nodes (batched) into A and resolve.

See appendix A1 (fractional degree) and A2 (reduced cost derivation).

Branch-and-bound

Each B&B node stores V_1 (fixed in) and V_0 (fixed out). At each node: apply bounds, run column generation, prune if LP bound cannot beat incumbent. If $\bar{x}$ integral, accept as a candidate; else branch.

Branching rule. Classify a fractional node by $d_{\mathrm{frac}}(v) = \sum_{u \in N(v) \cap A} \bar{x}_u$ versus 2λ.

- $d_{\mathrm{frac}}(v) < 2\lambda$: "hanging", branch zero-first.
- $d_{\mathrm{frac}}(v) \geq 2\lambda$: "core", branch one-first.

2λ is the marginal pair-penalty rate. Root fixes $x_q = 1$.

See appendix A3.

Triangle cuts

Problem. LP relaxation can spread fractional x mass across nodes while keeping pair variables w_{ij} at zero (e.g., $x_u = x_v = 0.5$ and $w_{uv} = 0$).

Solution. Add constraints that forbid this pattern:

$$x_u + x_v + x_w - w_{uv} - w_{vw} - w_{uw} \leq 1.$$

Note

- separated only on fractional nodes
- bounded number per round
- skipped if LP gap is small

κ-connectivity cuts

Lazy enforcement. On each integer candidate S, scan $t \in S \setminus \{q\}$ via $q \to t$ max-flow on the unit-capacity undirected support graph.

At the first t with flow $< \kappa$:

- $R =$ source-side of min-cut (residual-reachable from q)
- add cut inequality

$$\sum_{(u,v)\in\delta(R)} y_{uv} \;\geq\; \kappa\, x_t$$

  where $\delta(R) = \{(u, v) \in E : \text{exactly one of } u, v \in R\}$
- re-solve

One cut per round; the next violation, if any, is caught on re-solve.

See appendix A4.

Practical considerations

Early stopping

- soft time limit: no-improvement cap, fires only when an incumbent exists
- hard time limit: absolute wall-clock cap, fires regardless
- node limit: cap branch-and-bound nodes per subproblem
- gap tolerance: terminate when LP-incumbent gap drops below threshold

Lazy frontier materialization

- fetch unqueried frontier adjacencies only when solo pricing finds no improving column

With any limit active, the solver returns the best incumbent. The LP bound stays a valid upper envelope on the parametric optimum.

BFS baseline

Problem. Topology-only frontier expansion from q.

Algorithm. Layered breadth-first expansion from the query: each dequeued node is oracle-queried, its predecessors and successors enter the pool. Depth d bounds the number of layers explored.

vs BP

- no density objective: returns the whole frontier
- no size constraint $|S| \geq k$
- no κ-connectivity constraint

Cheapest baseline: one oracle query per visited node, no flow, no LP. Output size grows with d; comparable per-k analysis needs the grow-to-k post-processing.

Average-degree baseline

Problem

$$\max_{\substack{S \ni q \\ S \subseteq V_{\mathrm{loc}}}} \frac{|E(S)|}{|S|}$$

where V_{loc} is a BFS-bounded local subgraph around q.

vs BP

- objective: avg degree, not pair density (the two are not monotone in each other)
- no size constraint $|S| \geq k$
- no κ-connectivity constraint

Algorithm: adapted Goldberg's s–t max-flow algorithm (polynomial time)

- materializing the local subgraph up-front
- forcing $q \in S$ by keeping q on the source side of the cut

Grow-to-k

Equalizes baseline output cardinality with BP for per-k comparison.

Rule. Seed S, pool $P \supseteq S$. Repeat

$$v^* = \arg\max_{v \in P \setminus S} \left| N(v) \cap S \right|$$

until $|S| = k$ (or P is exhausted).

- BFS: seed $= \{q\}$, depth d becomes a minimum; expansion continues until the pool reaches k.
- AvgDeg: seed $= S^*$, fires only when $|S^*| < k$.

Greedy heuristic. Densest-at-least-k is NP-hard; no approximation guarantee.

Experiments

Exp 1: solver correctness on enumerable graphs

Erdős–Rényi $G(n = 20, p)$ for $p \in \{0.25, 0.50, 0.75\}$, seeds 0..4 (15 graphs). Brute-force enumerates every $S \ni q$.

Result on the grid $(k, \kappa) \in \{3, 4, 5\} \times \{0, 1, 2\}$

- AvgDeg matches BF optimum on 15/15 graphs.
- BP returned S satisfies $|S| \geq k$ and rooted κ-edge-connectivity on 30/30 cells; edge density matches BF on every feasible cell.
- Zero solver errors.

See appendix A6 for protocol details.

Dataset, split, methods

Cora-ML (CitationFull)

- 2,995 nodes, 8,416 directed edges, 7 paper labels

Pool pure sources with out-degree ≥ 2

- val 402, test 402
- train pool 2,191 provides labels, never queried
- hard subset: 59 test nodes where depth-1 BFS is wrong

Methods

- BFS depth-1, AvgDeg (Goldberg), BP
- all grown to $|S| \geq k$ (unless not possible)
- sweep $k \in \{3, 4, 5\}$; BP also $\kappa \in \{0, 1, 2\}$

Exp 2: cluster quality on val + test

Cora-ML val + test (804 queries), single seed.

Method	$\lvert S\rvert$	λ_2	ncut	avg.deg	dens.
$k = 3$					
BFS d=1	4.87 ± 2.75	0.96 ± 0.30	1.23 ± 0.15	1.17 ± 0.50	0.35 ± 0.11
AvgDeg	21.20 ± 15.20	0.32 ± 0.38	0.70 ± 0.53	**3.02 ± 1.55**	0.21 ± 0.12
BP κ=0	3.34 ± 0.71	1.23 ± 0.34	1.36 ± 0.26	1.20 ± 0.42	**0.51 ± 0.10**
BP κ=1	3.32 ± 0.67	1.25 ± 0.28	1.39 ± 0.13	1.19 ± 0.41	0.51 ± 0.10
BP κ=2	3.28 ± 0.72	**1.27 ± 0.27**	**1.41 ± 0.11**	1.10 ± 0.44	0.48 ± 0.12
$k = 4$					
BFS d=1	5.20 ± 2.56	0.78 ± 0.25	1.17 ± 0.09	1.22 ± 0.47	0.32 ± 0.10
AvgDeg	21.21 ± 15.18	0.31 ± 0.35	0.69 ± 0.53	3.02 ± 1.55	0.21 ± 0.11
BP κ=0	4.35 ± 0.78	0.75 ± 0.48	0.94 ± 0.53	1.59 ± 0.48	0.47 ± 0.09
BP κ=1	4.23 ± 0.76	0.92 ± 0.28	1.21 ± 0.10	1.47 ± 0.42	0.46 ± 0.09
BP κ=2	4.19 ± 0.72	0.94 ± 0.28	1.24 ± 0.08	1.34 ± 0.46	0.42 ± 0.12
$k = 5$					
BFS d=1	5.72 ± 2.36	0.66 ± 0.24	1.13 ± 0.08	1.29 ± 0.46	0.29 ± 0.09
AvgDeg	21.25 ± 15.14	0.30 ± 0.34	0.69 ± 0.52	3.02 ± 1.55	0.20 ± 0.11
BP κ=0	5.03 ± 0.72	0.50 ± 0.44	0.74 ± 0.57	1.76 ± 0.51	0.43 ± 0.09
BP κ=1	5.14 ± 1.03	0.73 ± 0.27	1.14 ± 0.08	1.67 ± 0.47	0.41 ± 0.09
BP κ=2	5.16 ± 1.13	0.72 ± 0.28	1.16 ± 0.08	1.52 ± 0.54	0.37 ± 0.11

Section best per column (higher better, $\lvert S\rvert$ excluded); **overall best** across k.

Exp 2: cost on val + test

$n = 804$ queries.

Method	wall (s)	queries	BB nodes	LP solves	cuts	t_{LP} (s)
$k = 3$						
BFS d=1	0.03 ± 0.01	5 ± 3	–	–	–	–
AvgDeg	0.08 ± 0.05	202 ± 210	–	–	–	–
BP κ=0	1.54 ± 15.19	179 ± 201	121 ± 837	131 ± 861	87 ± 315	0.93 ± 11.50
BP κ=1	1.47 ± 13.71	179 ± 201	121 ± 772	132 ± 796	90 ± 308	0.87 ± 10.4
BP κ=2	10.3 ± 53.1	180 ± 202	81 ± 429	1106 ± 5876	1106 ± 5868	6.26 ± 36.3
$k = 4$						
BFS d=1	0.03 ± 0.01	5 ± 3	–	–	–	–
AvgDeg	0.08 ± 0.05	202 ± 210	–	–	–	–
BP κ=0	9.85 ± 38.12	200 ± 210	833 ± 2025	881 ± 2079	492 ± 677	9.21 ± 36.31
BP κ=1	28.2 ± 79.7	200 ± 210	1549 ± 3593	1618 ± 3696	756 ± 1348	26.9 ± 76.8
BP κ=2	47.9 ± 103.3	200 ± 210	1363 ± 3061	3794 ± 9691	3300 ± 9483	44.1 ± 95.6
$k = 5$						
BFS d=1	0.04 ± 0.01	6 ± 2	–	–	–	–
AvgDeg	0.08 ± 0.05	202 ± 210	–	–	–	–
BP κ=0	66.7 ± 110.0	202 ± 210	2967 ± 4275	3079 ± 4392	1291 ± 1477	63.2 ± 105.6
BP κ=1	92.4 ± 130.6	202 ± 210	3799 ± 5311	3961 ± 5485	1992 ± 2825	88.4 ± 126.3
BP κ=2	96.0 ± 131.8	202 ± 210	3197 ± 4493	4409 ± 7318	3326 ± 6744	90.8 ± 126.7

BP soft cap 60 s, hard cap 300 s.

Exp 3: scope and prediction rule

- extrinsic perspective
- tune on val, evaluate on test and on the BFS-hard test subset

For each val or test query q:

- retrieve a community around q from the chosen method
- predict q's label by a uniform majority vote over the training-labelled members of that community
- ties broken by ascending label so the result is reproducible
- fallback (rare):
  - walk outward by BFS from q, forbidding val and test nodes as stepping stones, until a training neighbour is reached
  - if no training label is reachable, use the global training-majority label

Exp 3: tuning on val

Cora-ML val (402 queries). Mean $\pm$ bootstrap std.

Method	P	R	F1
$k = 3$			
BFS $d = 1$	0.836 $\pm$ 0.019	0.832 $\pm$ 0.020	0.826 $\pm$ 0.021
AvgDeg	0.714 $\pm$ 0.026	0.540 $\pm$ 0.024	0.558 $\pm$ 0.026
BP $\kappa = 0$	0.797 $\pm$ 0.021	0.787 $\pm$ 0.022	0.782 $\pm$ 0.022
BP $\kappa = 1$	0.799 $\pm$ 0.020	0.790 $\pm$ 0.021	0.785 $\pm$ 0.021
BP $\kappa = 2$	0.801 $\pm$ 0.020	0.794 $\pm$ 0.021	0.785 $\pm$ 0.022
$k = 4$			
BFS $d = 1$	**0.844 $\pm$ 0.019**	**0.843 $\pm$ 0.019**	**0.838 $\pm$ 0.020**
AvgDeg	0.717 $\pm$ 0.026	0.544 $\pm$ 0.024	0.563 $\pm$ 0.025
BP $\kappa = 0$	0.762 $\pm$ 0.021	0.683 $\pm$ 0.025	0.694 $\pm$ 0.025
BP $\kappa = 1$	0.808 $\pm$ 0.020	0.789 $\pm$ 0.021	0.789 $\pm$ 0.022
BP $\kappa = 2$	0.827 $\pm$ 0.020	0.810 $\pm$ 0.021	0.811 $\pm$ 0.021
$k = 5$			
BFS $d = 1$	0.814 $\pm$ 0.021	0.816 $\pm$ 0.021	0.810 $\pm$ 0.021
AvgDeg	0.717 $\pm$ 0.026	0.544 $\pm$ 0.024	0.563 $\pm$ 0.025
BP $\kappa = 0$	0.725 $\pm$ 0.023	0.616 $\pm$ 0.026	0.628 $\pm$ 0.026
BP $\kappa = 1$	0.772 $\pm$ 0.024	0.744 $\pm$ 0.023	0.741 $\pm$ 0.024
BP $\kappa = 2$	0.795 $\pm$ 0.022	0.784 $\pm$ 0.021	0.780 $\pm$ 0.022

Section best (max per column within k); **overall best** (max across k).

Exp 3: evaluation on test

Per-method val-best applied to Cora-ML test (402 queries). Mean $\pm$ bootstrap std.

Method	val-best params	P	R	F1
BFS	$d = 1,\ k = 4$	**0.871 ± 0.016**	**0.853 ± 0.020**	**0.856 ± 0.019**
AvgDeg	$k = 4$	0.772 ± 0.022	0.579 ± 0.024	0.603 ± 0.024
BP	$k = 4,\ \kappa = 2$	0.818 ± 0.021	0.815 ± 0.021	0.811 ± 0.021

Exp 3: evaluation on the hard subset

59 Cora-ML test nodes where the plain depth-1 BFS vote is wrong. Mean $\pm$ bootstrap std.

Method	val-best params	P	R	F1
BFS	$d = 1,\ k = 4$	0.183 ± 0.061	0.097 ± 0.049	0.080 ± 0.042
AvgDeg	$k = 4$	$\mathbf{0.269 \pm 0.084}$	0.251 ± 0.079	0.146 ± 0.056
BP	$k = 4,\ \kappa = 2$	0.174 ± 0.053	$\mathbf{0.282 \pm 0.090}$	$\mathbf{0.190 \pm 0.056}$

Bold = max per column.

OpenAlex frontend

Same solver, live oracle

- Solver is oracle-agnostic; swap simulation client for an OpenAlex client.
- Each oracle call fetches one paper's outgoing references and capped incoming citations.
- Nodes are materialized lazily; API cost scales with what BP actually pulls in.
- Density, κ-connectivity, grow-to-k behaviour unchanged.

Conclusion

Main interpretation

Cora-ML; cluster quality and cost on $n = 804$ (val + test); test F1 on $n = 402$; hard subset on $n = 59$.

- **BP wins internal geometry at every** k: $\mathrm{BP}_{\kappa=2}$ leads λ_2 and ncut, $\mathrm{BP}_{\kappa=0}$ leads edge density.
- **AvgDeg picks the densest set by avg degree**, but via much larger sets ($|S| \approx 21$); per-pair edge density drops.
- **Same-k test F1**: BFS leads (0.856), BP runs second (0.811), AvgDeg trails (0.603) as its large sets dilute the majority vote.
- **Hard subset**: BFS near-zero by construction (0.080), AvgDeg middle (0.146), BP leads (0.190).
- **Cost**: BFS and AvgDeg fast; BP grows steeply with k and κ.

Claim. BP is a dense-community retriever; classification F1 is a downstream readout of the retrieved geometry.

Limitations

- Numbers only cover Cora_ML; more datasets in progress.
- Temporal split makes one-hop BFS match the homophily structure: test nodes are pure sources whose direct citations sit in the train set. The hard subset is a conditional diagnostic on queries where the plain depth-1 BFS majority vote (no grow-to-k) is wrong, so the BFS solver scores near-zero there by construction.
- Exact optimality is over the discovered graph region; the oracle interface bounds what BP can reason about.
- Solver limits (time, branch-and-bound nodes, Dinkelbach / column-generation iterations) return possibly-suboptimal incumbents on the tail.

Next steps

- Tighter LP relaxation
- Inverse-out-degree edge weights
- Warm-start with densest-subgraph incumbent
- Incorporate semantic information

Thank you.

A1. Fractional degree: directionality and cap

Directionality

- $N(v) = N^+(v) \cup N^-(v)$: undirected union of outgoing references and incoming citations
- reciprocal pairs $(u, v), (v, u) \in E$ contribute v twice, matching directed edge density

Robustness to the incoming-citation cap

- every active node is oracle-queried, so its complete outgoing list is recorded
- ingestion is symmetric: writing $u \to v$ also marks $v \in \text{adj_in}[u]$ and $u \in \text{adj_in}[v]$
- every edge with at least one endpoint in A is captured, regardless of the cap

Cap limits frontier coverage (which papers can ever join $A \cup F$), not the correctness of $d_{\text{frac}}(v) \cap A$.

A2.1. Reduced cost: setup

RMP at parameter λ, active set A:

$$\begin{aligned}
\max \quad & \sum_{(u,v)\in E(A)} y_{uv} - 2\lambda \sum_{\substack{u<v \\ u,v\in A}} w_{uv} \\
\text{s.t.} \quad & x_q = 1 \\
& \sum_{u\in A} x_u \geq k \quad (\text{dual: } \pi) \\
& y_{uv} \leq x_u,\ y_{uv} \leq x_v \\
& w_{uv} \geq x_u + x_v - 1 \\
& x, y, w \in [0,1]
\end{aligned}$$

LP optimum: $\bar{x}, \bar{y}, \bar{w}$. Size dual: π.

Question. If $f \in F$ joins A, does the LP improve?

A2.2. Reduced cost: probe

Pretend f joins A. New variables enter the LP:

- $x_f = 1$ (probe value)
- y_{fv} for each edge $(f, v) \in E$ with $v \in A$
- w_{fv} for each $v \in A$ (every pair, edge or not)

Freeze $\bar{x}_v$ for $v \in A$ at current LP values. Optimize new variables freely. The resulting objective change is the reduced cost:

$$\bar{c}_f \;=\; \underbrace{(y\text{-gain})}_{\text{Step 1}} + \underbrace{(w\text{-penalty})}_{\text{Step 2}} + \underbrace{(\text{size correction})}_{\text{Step 3}}.$$

A2.3. Reduced cost step 1: edge gain (y terms)

For each new edge $(f, v) \in E$ with $v \in A$:

Bounds.

$$y_{fv} \leq \min(x_f, x_v) = \min(1, \bar{x}_v) = \bar{x}_v.$$

Objective coefficient. $+1$.

Max problem, positive coefficient $\Rightarrow$ upper bound binds:

$$y^\star_{fv} = \bar{x}_v.$$

Total y-contribution:

$$\sum_{v \in N(f) \cap A} \bar{x}_v \;=\; d_{\mathrm{frac}}(f).$$

A2.4. Reduced cost step 2: pair penalty (w terms)

For each $v \in A$ (every pair, edge or not):

Bound.

$$w_{fv} \geq x_f + x_v - 1 = \bar{x}_v.$$

Objective coefficient. -2λ.

Max problem, negative coefficient $\Rightarrow$ lower bound binds:

$$w^\star_{fv} = \bar{x}_v.$$

Total w-contribution:

$$-2\lambda \sum_{v \in A} \bar{x}_v.$$

A2.5. Reduced cost step 3: size dual (π term)

Size constraint $\sum_{u \in A} x_u \geq k$ has dual price π.

Dual interpretation. π is the marginal LP value of relaxing k by one unit.

When $x_f = 1$, the LHS grows by one. The reduced cost picks up a shadow term $-\pi$.

- Size constraint slack at LP optimum: $\pi = 0$, no contribution.
- Size constraint binding ($|S| = k$): π takes the sign that makes $-\pi$ the correct correction.

A2.6. Reduced cost: sum and decision

Sum the three contributions:

$$\bar{c}_f \;=\; d_{\mathrm{frac}}(f) \;-\; 2\lambda \sum_{v \in A} \bar{x}_v \;-\; \pi.$$

Decision rule

- $\bar{c}_f > 0$: adding f strictly improves the LP $\Rightarrow$ promote to A.
- $\bar{c}_f \leq 0$: no gain $\Rightarrow$ keep in F.
- $\max_{f \in F} \bar{c}_f \leq 0$: no frontier node helps. Column generation terminates and the RMP LP objective is a valid upper bound on the parametric subproblem.

A3. Branching threshold 2λ

Dinkelbach objective:

$$\sum_{(u,v)\in E} y_{uv} - 2\lambda \sum_{u<v} w_{uv}.$$

Each pair carries penalty rate 2λ.

Marginal contribution of v at fractional rate (per active neighbor u):

- edge gain: $+\bar{x}_u$ if $(u, v) \in E$
- pair penalty: $-2\lambda\bar{x}_u$ (every $u \in A$)

Aggregate: v pays off when $d_{\mathrm{frac}}(v) \geq 2\lambda$.

- $d_{\mathrm{frac}}(v) \geq 2\lambda$: edge gain covers pair penalty rate $\Rightarrow$ core-like, branch one-first
- $d_{\mathrm{frac}}(v) < 2\lambda$: penalty dominates $\Rightarrow$ hanging, branch zero-first

Heuristic threshold; at LP optimum, included nodes typically satisfy $d_{\mathrm{frac}}(v) \geq 2\lambda$.

A4. κ-cut inequality: derivation

Menger's theorem (undirected). For $t \in S$,

$$\max(\text{edge-disjoint } q\text{-}t \text{ paths in } E(S)) \;=\; \min_{\substack{R \ni q \\ R \not\ni t}} |\delta(R) \cap E(S)|.$$

So S has $\geq \kappa$ edge-disjoint q-t paths $\iff$ every q-t cut R has $\geq \kappa$ internal edges.

LP encoding. $y_{uv} = 1$ iff edge (u, v) internal to S.

- $x_t = 1$: need $\sum_{(u,v)\in\delta(R)} y_{uv} \geq \kappa$
- $x_t = 0$: no requirement

Combine:

$$\sum_{(u,v)\in\delta(R)} y_{uv} \;\geq\; \kappa\, x_t.$$

Validity. Holds at every κ-edge-connected integer S, so no feasible solution is excluded.

A5.1. Nonlinear fractional 0-1 program

Let $x_v \in \{0,1\}$ indicate $v \in S$ and $y_{uv} \in \{0,1\}$ indicate edge (u,v) is internal:

$$\begin{aligned}
\max \quad & \frac{\sum_{(u,v)\in E} y_{uv}}{\left(\sum_{u\in V} x_u\right)\left(\sum_{u\in V} x_u - 1\right)} \\
\text{s.t.} \quad & x_q = 1 \\
& \sum_{u\in V} x_u \geq k \\
& y_{uv} \leq x_u, \quad y_{uv} \leq x_v \quad \forall (u,v) \in E
\end{aligned}$$

The lower bound $y_{uv} \geq x_u + x_v - 1$ is not needed because y has positive objective coefficient.

A5.2. Dinkelbach parametric subproblem (bilinear form)

At parameter λ:

$$
\begin{aligned}
Z(\lambda) = \max \quad & \sum_{(u,v)\in E} y_{uv} - \lambda \left[\left(\sum_{u\in V} x_u \right)^2 - \sum_{u\in V} x_u \right] \\
\text{s.t.} \quad & x_q = 1 \\
& \sum_{u\in V} x_u \geq k \\
& y_{uv} \leq x_u, \quad y_{uv} \leq x_v \quad \forall (u,v) \in E \\
& x \in \{0,1\}^{|V|},\ y \in \{0,1\}^{|E|}
\end{aligned}
$$

A5.3. Linearizing the penalty

$|S|(|S|-1) = (\sum_v x_v)^2 - \sum_v x_v$. For binary x, $x_v^2 = x_v$, so

$$|S|(|S|-1) = 2\sum_{u<v} x_u x_v.$$

Introduce pair variable $w_{uv} \approx x_u x_v$ with $w_{uv} \geq x_u + x_v - 1$. Since w has negative coefficient, the lower bound is enough at integer solutions.

A5.4. Linearized parametric subproblem

At parameter λ:

$$
\begin{aligned}
Z(\lambda) = \max \quad & \sum_{(u,v)\in E} y_{uv} - 2\lambda \sum_{u<v} w_{uv} \\
\text{s.t.} \quad & x_q = 1 \\
& \sum_{u\in V} x_u \geq k \\
& y_{uv} \leq x_u, \quad y_{uv} \leq x_v \quad \forall (u,v) \in E \\
& w_{uv} \geq x_u + x_v - 1 \quad \forall u < v \\
& x \in \{0,1\}^{|V|},\ y \in \{0,1\}^{|E|},\ w \in \{0,1\}^{\binom{|V|}{2}}
\end{aligned}
$$

A5.5. Restricted Master Problem

For active set A:

$$\begin{aligned}
\max \quad & \sum_{(u,v)\in E(A)} y_{uv} - 2\lambda \sum_{\substack{u,v\in A\\ u<v}} w_{uv} \\
\text{s.t.} \quad & x_q = 1 \\
& \sum_{u\in A} x_u \geq k \quad \text{(dual: } \pi\text{)} \\
& y_{uv} \leq x_u, \quad y_{uv} \leq x_v \quad \forall (u,v) \in E(A) \\
& w_{uv} \geq x_u + x_v - 1 \quad \forall u < v \in A \\
& 0 \leq x, y, w \leq 1
\end{aligned}$$

A5.6. Pruning integer incumbents

Since $|S| \geq k$, integer solutions may exceed k. Greedily remove nodes whose removal strictly improves the parametric objective.

- candidate: node with the smallest internal degree (weakest edge contribution)
- never remove q
- stop at $|S| = k$ or when no single removal helps
- preserve connectivity when $\kappa > 0$

Effect. Stronger incumbent, smaller returned community.

A6. Exp 1 protocol detail

Brute-force oracle

- enumerate every $S \ni q$ with $|S| \geq k$
- for each (k, κ), record the maximum-density S with $\lambda(S) \geq \kappa$
- mark cells with no feasible S as infeasible

Solver

- run on every (method, k, κ, seed) cell
- no time, node, gap, or Dinkelbach limit
- skip cells the brute force flagged infeasible (solver otherwise exhausts B&B proving infeasibility; wall time blows up on sparse seeds)